Devashree Gupta

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Dept. of Biomedical & Electrical Engineering
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Summary

Fourth-year undergraduate studying Biomedical Engineering and Electrical Engineering, graduating May 2027. Interested in pursuing a career in medical devices, with a passion for neuroengineering and reproductive healthcare. Experience in sensor characterization, mixed-signal circuit design, embedded systems, and signal processing.

Education

Johns Hopkins University Baltimore, MD
B.S., Biomedical Engineering & B.S., Electrical Engineering (Dual Degree) Aug. 2023 – May 2027
(Expected)
Cumulative GPA: 3.68
Relevant Coursework: Control Systems, Nonlinear Dynamics, FPGA Programming, Microfabrication Lab, Biological Models and Simulations, Digital Signal Processing, VLSI, Advanced Electronics Design

Research

Department of Biomedical Engineering, Johns Hopkins University *July 2025 – June 2026*
Research Assistant, Engineering *Baltimore, MD*

- Led electrical design of mosquito traps for attraction, capture, powder marking, and release for surveillance of *Aedes aegypti* mosquito populations in Brazil, Florida, and Costa Rica.
- Designed a relay control PCB in KiCAD and developed an Arduino-based embedded system for automated nightly operation.
- Isolated repeated field failures of the trap visual system to insufficient illumination and re-designed lighting to improve image quality.

Comillas Pontifical University *May 2025 – Aug. 2025*
Signal Processing Engineering Intern *Madrid, Spain*

- Built an RF signal-processing and classification pipeline to detect and identify drones by their RF signatures.
- Denoised raw RF captures with FFT-based filtering and wavelet transforms, improving SNR by 40%.
- Trained a convolutional neural network on the preprocessed signatures, reaching 88% classification accuracy.

Brain Value Ltd. *July 2024 – Sept. 2024*
Data Science Intern *Bangalore, India*

- Collected, processed, and analyzed biomarker and clinical trial datasets using machine learning methods; prepared reports and visualizations to characterize biomarker relationships.

- Processed clinical trial data, applying regression analysis, clustering, and predictive modeling using scikit-learn.

Industry Experience

Medtronic, Electrosurgical Generator Team
June 2026 – Aug. 2026
Sensor Hardware Engineering Intern *Lafayette, CO*

- Characterized the voltage and current sensing of a next-generation surgical generator across its full operating range to establish software calibration baselines, presented findings to 300+ Medtronic employees.
- Automated the full characterization loop in Python, sweeping loads, PWM drive modes, and harmonics on a self-built bench mount.
- Defined sensor gain and phase values from characterized data, delivering calibration baselines used in production firmware.
- Root-caused discrepancies between external probe and onboard measurements to unaccounted parasitic capacitance of traces.

Johnson & Johnson MedTech, ECMO Team
Jan. 2026 – June 2026
Product Development Engineering Intern *Baltimore, MD*

- Developed test methods for cavitation and air handling in blood pumps, supporting next-generation pump design and IDE submission to the FDA, presented projects and findings to 100+ employees.
- Built end-to-end cavitation test protocol in accordance with ISO 18242, using a high-speed camera to detect cavitation formation at the impeller inlet, enabling repeatable compliance testing across builds.
- Investigated a de-airing method for blood pumps to prevent gas embolism, identifying air localization and modeling design changes for optimal air removal, documented findings in the DHF.
- Drafted and executed Test Method Validation protocols and tool specifications for critical components, applying Gage R&R and capability analysis using Minitab and a Keyence IM.

Projects

Airway Adaptor for Robotic Bronchoscopies

- Led prototyping for a robotic bronchoscopy attachment to keep visualization of the lungs unobstructed during multi-tool endotracheal intubation, allowing interventional pulmonologists to detect pulmonary hemorrhage early.
- Collaborated in an 8-person team, analyzed 500+ clinical studies, and shadowed pulmonologists in the operating room to derive user needs and design requirements.
- Modeled the device assembly in SolidWorks and fabricated 20+ design iterations by resin 3D printing, successfully testing the device in a mock OR.

Transperineal Biopsy Collection System

- Led verification and validation of a new transperineal biopsy system to improve prostate cancer diagnosis.

- Designed and built LVDT-instrumented fixtures to measure needle firing velocity, establishing the verification method for the firing mechanism's core design requirement.
- Completed a working biopsy-gun prototype designed against FDA and ISO requirements and validated it through a progression of low- to high-fidelity test methods.

CMOS Photosensor
- Designed a 2.5 mm × 2.5 mm CMOS image sensor in Cadence Virtuoso from schematic to verified layout.
- Integrated 3-transistor active pixel sensors with photodiodes, row shift registers, and column scan logic.
- Verified the completed layout with full DRC and LVS checks.

Password-Encrypted Digital Keypad
- Built a passcode-locked keypad in discrete logic that unlocks only on a correct 7-digit entry sequence.
- Architected a Moore finite state machine with one-hot encoding across 8 sequential passcode-validation states.
- Prototyped in hardware with D flip-flops, logic gates, and a clock circuit, adding pull-down resistors to eliminate floating inputs and LED indicators for entry feedback.

LED Music Sync Circuit
- Designed an analog circuit that drives LEDs in sync with the frequency content of microphone audio, using three frequency bands.
- Unit-tested the electret-microphone amplification and filtering stages on a breadboard, then verified performance and tuned the frequency response in LTSpice simulation.

BRAF V600E Mutation Study
- Investigated how the BRAF V600E mutation alters MAPK/ERK signaling to drive uncontrolled proliferation.
- Extracted cellular metrics with CellProfiler, then compared oncogenic and wild-type populations in MATLAB using heatmaps, line graphs, and statistical analyses.
- Found significantly higher ERK activity, more cytoplasmic localization, and smaller nuclei in oncogenic cells.

Teaching

Johns Hopkins University *Jan. 2025 – Dec. 2025*
Physics Learning Assistant *Baltimore, MD*
- Reinforced electricity and magnetism concepts by leading group discussion and problem-solving sessions, in collaboration with professors and teaching assistants.

Johns Hopkins University *Aug. 2024 – Dec. 2025*
First-Year Mentor *Baltimore, MD*
- Supported incoming students via orientation activities, group events, and academic and campus resources.

Johns Hopkins University
Design Cornerstone Teaching Assistant *Baltimore, MD*

- Mentored first-year engineering teams through hands-on design and prototyping projects, advising on multidisciplinary design processes, electronics, Arduino hardware/software, and 3D printing.
- Helped troubleshoot student design challenges, provided feedback on prototypes and presentations, and collaborated with faculty to support course activities and maintain a safe prototyping environment.

Technical Skills

Programming & Data Analysis: Python (NumPy, SciPy, pandas, scikit-learn), C/C++, MATLAB, LabVIEW, NI DAQ, SQL, Git, Minitab
Circuits & Embedded Systems: KiCAD, LTSpice, Cadence Virtuoso, Verilog, Arduino
CAD & Prototyping: SolidWorks (incl. FEA Simulation), Rapid Prototyping, 3D Printing (SLA/FDM), Machining (CNC, Lathe, Weld, Mill, Laser Cutting)

Leadership & Service

AKDPhi — *Professional Development Chair*: Hosted resume workshops, networking events, and a headshot event.
AKDPhi — *Fundraising Chair*: Raised $3,000 for breast cancer awareness through fundraising activities.
Johns Hopkins Outpatient Center — *Community Connection Advocate* (Jan. 2024 – May 2025): Assisted underserved patients and families in overcoming barriers to social determinants of health.
Believe in Art — *Volunteer*: Volunteered at the Keswick Multi-Care Center, spending time and crafting with older patients living with dementia.

Languages

English, Hindi, French, Marathi, Spanish